Conformal Risk Control for Safety-Critical Wildfire Evacuation Mapping: A Comparative Study of Tabular, Spatial, and Graph-Based Models

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Abstract

Every wildfire prediction model deployed today shares a dangerous property: none of them guarantee how many fires they will miss. Despite extensive work on wildfire spread prediction using deep learning, no prior study has applied distribution-free safety guarantees to this domain—leaving evacuation planners reliant on probability thresholds with no formal assurance. We address this gap by presenting, to our knowledge, the first application of conformal risk control (CRC) (Angelopoulos et al., 2022) to wildfire spread prediction, providing finite-sample guarantees on false negative rate ($\mathrm{FNR} \leq 0.05$). We expose a stark failure: across three model families of increasing complexity—tabular (LightGBM, AUROC 0.854), convolutional (Tiny U-Net, AUROC 0.969), and graph-based (Hybrid ResGNN-UNet, AUROC 0.951)—standard thresholds capture only 7–52% of true fire pixels. CRC eliminates this failure uniformly. Our central finding is that *model architecture determines evacuation efficiency, while CRC determines safety*: a 470K-parameter U-Net with CRC achieves 94.7% fire coverage while flagging only 14.9% of pixels—4.2× more efficient than LightGBM—while the architecturally more complex 229K-parameter graph model achieves lower AUROC. We further propose a three-way CRC extension producing SAFE/MONITOR/EVACUATE zones for operational triage, and characterize a fundamental limitation of prevalence-weighted bounds under extreme class imbalance ($\sim$5% fire prevalence). All models, calibration code, and evaluation pipelines are released for reproducibility.

1. Introduction

Wildfire evacuation planning requires prediction systems that prioritize safety over accuracy. In this domain, a false negative—a fire pixel classified as safe—poses direct risk to human life, while false positives carry their own operational cost: overly broad evacuation perimeters trigger *shadow evacuations*, where residents outside the official zone self-evacuate, congesting road networks and potentially trapping those in genuine danger (Zhao et al., 2022). This dual asymmetry renders standard accuracy-based evaluation inadequate: a system reporting 90% pixel-level accuracy may simultaneously miss the majority of fire pixels due to extreme class imbalance, as we demonstrate empirically in this work.

Despite decades of wildfire prediction research (Jain et al., 2020) and the recent availability of distribution-free safety tools such as *conformal risk control* (CRC) (Angelopoulos et al., 2022), no prior work has applied CRC—or any conformal method—to wildfire spread prediction. Wildfire ML continues to evaluate models on accuracy-based metrics (F1, IoU, AUROC) that are blind to the safety-critical nature of the task, while the conformal prediction community has focused on medical imaging, autonomous driving, and natural language processing. This leaves a clear gap: **the domain where false negatives are most dangerous has no formal safety guarantees.**

We bridge this gap. CRC selects a decision threshold from held-out calibration data such that $\mathbb{E}[\mathrm{FNR}] \leq \alpha$ on test data, regardless of the underlying model's quality or calibration (Angelopoulos & Bates, 2021). A natural question then motivates our study: *can a sufficiently powerful model eliminate the need for post-hoc safety calibration?* To answer this, we train and evaluate three architectures of increasing complexity on the Next Day Wildfire Spread (NDWS) dataset (Huot et al., 2022): **(i)** a tabular LightGBM baseline (Ke et al., 2017) treating each pixel independently, **(ii)** a Tiny U-Net (Ronneberger et al., 2015) capturing local spatial structure, and **(iii)** a Hybrid ResGNN-UNet combining convolutional encoders with graph attention layers (Veličković et al., 2018) to model fire propagation on pixel graphs. All models are trained from scratch on identical data splits.

Our experiments yield a clear answer: **no model is safe without CRC**. At their respective optimal thresholds, LightGBM misses 92.8% of fires, U-Net misses 61.0%, and the ResGNN-UNet misses 48.5%. With CRC, the simplest spatial model achieves 94.7% coverage while flagging only 14.9% of pixels—outperforming the architecturally more complex graph model in both safety and efficiency.

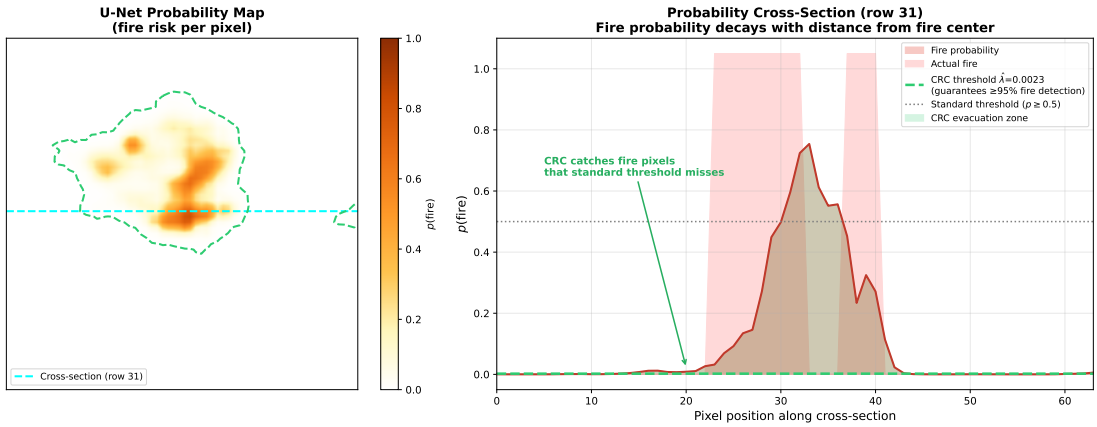

Figure 1. Why standard thresholds fail. A cross-section through a U-Net fire prediction shows probability peaking at $\sim$0.8 and decaying smoothly. The standard threshold ($\hat{p} \geq 0.5$, gray) misses the majority of the fire region; the CRC threshold ($\hat{\lambda} = 0.002$, green) captures it entirely.

Contributions.

- We demonstrate that standard probability thresholding is *catastrophically unsafe* for wildfire evacuation across three model families, with fire coverage ranging from 7% to 52%.
- We show that CRC provides guaranteed $\geq$95% fire coverage regardless of model quality, and that architecture determines *efficiency* (evacuation zone size)—with spatial models achieving 4.2$\times$ tighter zones than tabular baselines under the same safety guarantee.
- We propose a three-way CRC extension with SAFE/MONITOR/EVACUATE zones and characterize its behavior under extreme class imbalance, identifying a fundamental limitation of prevalence-weighted bounds when $\pi_0 \gg \pi_1$.
- We provide a complete, reproducible pipeline with deterministic splits and open-source code for all experimental phases.

2. Related Work

Machine learning for wildfire prediction. Machine learning has been extensively applied to wildfire risk assessment, spread prediction, and resource allocation (Jain et al., 2020). Deep learning approaches—including CNNs, recurrent architectures, and encoder-decoder models—have demonstrated strong performance on next-day spread prediction tasks (Radke et al., 2019). The NDWS dataset (Huot et al., 2022) established a standardized benchmark for pixel-level wildfire spread prediction from remote-sensing data, enabling systematic comparison across architectures. Graph neural networks (Kipf & Welling, 2017; Veličković et al., 2018) have been explored for modeling fire propagation dynamics, motivated by the intuition that fire spreads along spatial adjacency relations. However, *every* existing work in this domain evaluates exclusively on accuracy-based metrics (F1, IoU, AUROC). None provides formal safety guarantees on the false negative rate—the metric that determines whether an evacuation zone actually contains the fire.

Conformal prediction and distribution-free inference. Conformal prediction, introduced by Vovk et al. (2005) and surveyed by Shafer & Vovk (2008), provides distribution-free prediction sets with guaranteed finite-sample coverage under exchangeability. Romano et al. (2020) extended conformal methods to classification with adaptive prediction sets, and Barber et al. (2021) developed the Jackknife+ for predictive inference without sample splitting. Angelopoulos et al. (2022) generalized these ideas to *conformal risk control* (CRC), which controls arbitrary monotone loss functions—including the false negative rate loss central to our work. Angelopoulos & Bates (2021) provides an accessible treatment of the key ideas. CRC has since been applied to medical imaging, autonomous driving, and natural language processing—but not to any geospatial or environmental disaster domain. To our knowledge, **this is the first application of conformal risk control to wildfire spread prediction**, where the asymmetric cost structure (missed fires endanger lives; over-broad zones cause shadow evacuations and traffic gridlock (Zhao et al., 2022)) makes formal FNR guarantees uniquely valuable.

Uncertainty quantification and calibration. Guo et al. (2017) demonstrated that modern neural networks are poorly calibrated, with confidence scores that do not reflect true probabilities. Standard recalibration methods such as Platt scaling (Platt, 1999) and temperature scaling (Guo et al., 2017) improve calibration but provide no finite-sample coverage guarantees. Recent work has explored Bayesian and

ensemble-based uncertainty quantification for wildfire forecasting, but these methods produce heuristic uncertainty estimates without distribution-free guarantees. Our work shows that even well-calibrated models remain unsafe under standard thresholding due to the extreme class imbalance inherent in wildfire data ($\sim$5% fire pixels), motivating distribution-free methods that provide formal bounds rather than approximate confidence.

Safety-critical machine learning. Amodei et al. (2016) outlined fundamental challenges in deploying ML systems safely, including distributional shift and reward hacking. In safety-critical domains such as medical diagnosis and autonomous systems, the cost of false negatives far exceeds that of false positives, motivating methods that provide formal guarantees rather than empirical performance alone. Wildfire evacuation exemplifies this asymmetry more starkly than most domains: a missed fire pixel risks lives, while an over-broad perimeter causes cascading logistical failures in transportation networks (Zhao et al., 2022). Yet this is precisely the domain where formal safety guarantees have been absent. Our work closes this gap: we demonstrate that the difference between standard thresholds and CRC-calibrated thresholds corresponds to a 42–88 percentage-point improvement in fire detection rate.

3. Preliminaries

3.1. Problem Formulation

We consider pixel-level binary classification for wildfire spread prediction. Given an input sample $x \in \mathbb{R}^{H\times W\times C}$ (in our setting, $H{=}W{=}64$, $C{=}12$) and binary fire mask $y \in \{0,1\}^{H\times W}$, a model f_θ outputs per-pixel probability estimates $\hat{p}_{ij} = f_\theta(x)_{ij} \in [0,1]$. Binary predictions are obtained by thresholding:

$$\hat{y}_{ij}(\lambda) = \mathbb{1}\{\hat{p}_{ij} \geq \lambda\}. \tag{1}$$

Our primary safety objective is to control the false negative rate:

$$\mathrm{FNR}(\lambda) = \frac{\sum_{ij} y_{ij}(1-\hat{y}_{ij}(\lambda))}{\sum_{ij} y_{ij}} \leq \alpha, \quad \alpha = 0.05, \tag{2}$$

equivalently requiring empirical fire coverage $1 - \mathrm{FNR} \geq 95\%$.

3.2. Conformal Risk Control

CRC (Angelopoulos et al., 2022) provides a principled method for selecting λ with finite-sample guarantees. Given calibration examples $\{(x_i, y_i)\}_{i=1}^n$ drawn exchangeably with the test data, let m denote the number of positive (fire) calibration pixels and $p_{(1)} \leq \cdots \leq p_{(m)}$ their sorted predicted probabilities. CRC selects:

$$\hat{\lambda} = p_{(\lceil \alpha\,(m+1)\rceil)}, \tag{3}$$

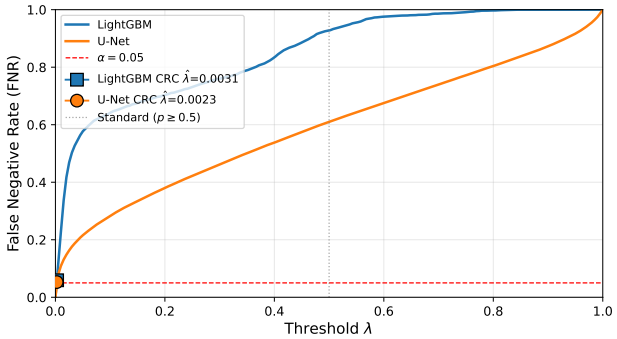

Figure 2. FNR as a function of decision threshold for LightGBM and U-Net. The red dashed line marks $\alpha = 0.05$. The U-Net maintains safe FNR over a much wider threshold range, explaining its 4.2× smaller evacuation zones under CRC. Vertical lines mark $\hat{\lambda}$.

the α-quantile of positive-pixel probabilities with finite-sample correction. At threshold $\hat{\lambda}$, at most $\lceil \alpha(m{+}1)\rceil - 1$ positive pixels fall below it, controlling the empirical FNR $\leq \alpha$. The resulting guarantee is:

$$\mathbb{E}[\mathrm{FNR}(\hat{\lambda})] \leq \alpha, \tag{4}$$

which holds *regardless* of the model's quality, calibration, or architecture—requiring only exchangeability of calibration and test data.

Safety–efficiency decomposition. A key insight motivating our study is that CRC *decouples* safety from efficiency. Equation (4) holds for any model; what varies is the *prediction set size*—the fraction of pixels flagged as fire. A model with strong fire/no-fire separation allows CRC to set a higher $\hat{\lambda}$, flagging fewer pixels while maintaining the same safety guarantee. A poorly separating model forces $\hat{\lambda}$ toward zero, expanding the evacuation zone. Thus, the model determines efficiency; CRC determines safety. Figure 2 illustrates this: the U-Net maintains safe FNR over a much wider threshold range than LightGBM, explaining the large difference in evacuation zone size when both are wrapped in CRC.

4. Methods

We train three architectures of increasing complexity on the same dataset and splits, then apply CRC uniformly.

4.1. Data

We use the NDWS dataset (Huot et al., 2022; Haddad, 2024): 18,545 samples of 64×64 patches with 12 input channels (elevation, weather, vegetation indices, population density, previous fire mask) and a binary fire target. We enforce a deterministic split (seed 42): **train** 12,981 (70%), **calibration** 2,781 (15%), **test** 2,783 (15%). The calibration set is never used during model training. Fire targets include

no-data pixels (-1); we construct valid-pixel masks and apply masked loss and evaluation throughout. Only $\sim$5% of valid pixels are labeled fire, creating extreme class imbalance with profound implications for threshold selection (Section 5).

4.2. Model 1: Tabular Baseline (LightGBM)

We flatten each 64×64 patch into pixel-level rows with 12 features and train an LGBMClassifier (Ke et al., 2017) with binary cross-entropy objective. To control memory, we subsample 5% of training pixels. This model treats each pixel independently and serves as a baseline to isolate the contribution of spatial inductive biases.

4.3. Model 2: Tiny U-Net

We implement a shallow U-Net (Ronneberger et al., 2015) with two encoder blocks ($12\to32\to64$ channels), a bottleneck ($64\to128$), and two decoder blocks with skip connections ($\sim$470K parameters). A final 1×1 convolution with sigmoid activation produces per-pixel probabilities. Training uses AdamW (lr= 10^{-3}, weight decay= 10^{-4}) with cosine annealing over 50 epochs, masked BCE loss, and best-model checkpointing on validation loss.

4.4. Model 3: Hybrid ResGNN-UNet

To test whether more expressive architectures improve probability separation, we implement a Hybrid ResGNN-UNet ($\sim$229K parameters): **(i)** a three-block CNN encoder ($12\to32\to64\to64$) downsampling to 8×8; **(ii)** a graph attention bottleneck with three GATConv layers (Veličković et al., 2018) (4 heads each) on an 8-connected pixel grid, with a residual connection; **(iii)** a U-Net decoder with three upsampling blocks, skip connections, and Dropout2d. Training uses AdamW (lr= 3×10^{-4}, weight decay= 10^{-3}) with BCEWithLogitsLoss (pos_weight= 8.0), cosine annealing over 25 epochs, gradient clipping, and early stopping (patience=7).

The GNN bottleneck treats the downsampled feature map as a graph where each pixel attends to its 8 spatial neighbors, designed to capture fire propagation patterns.

4.5. CRC Calibration

For each model, we compute per-pixel probabilities on the calibration set, then select $\hat{\lambda}$ via Equation (3) with $\alpha = 0.05$. This threshold is applied unchanged to the test set.

4.6. Three-Way CRC Extension

We extend standard CRC to a three-way decision framework that routes each pixel to one of three actions:

$$\begin{aligned} \text{SAFE} &: \hat{p} < \lambda_{\min}, \\ \text{MONITOR} &: \lambda_{\min} \le \hat{p} < \lambda_{\max}, \\ \text{EVACUATE} &: \hat{p} \ge \lambda_{\max}. \end{aligned} \tag{5}$$

The MONITOR zone captures epistemic uncertainty and routes ambiguous pixels to human review, rather than forcing a binary automated decision.

Threshold selection uses asymmetric cost penalties (c_{fn}=5, c_{fp}=1), reflecting the operational reality that a missed fire pixel risks lives whereas an over-broad perimeter causes logistical disruption but not direct harm (Zhao et al., 2022). The shift interval $\rho \in [0.9, 1.1]$ accounts for mild prevalence variation between calibration and deployment seasons; wider intervals (e.g., $[0.5, 2.0]$) cause the safety correction to exceed α, making calibration infeasible under $\sim$5% fire prevalence. The prevalence-weighted bound $B_{\text{pw}} = \max(c_{\text{fn}}\pi_1, c_{\text{fp}}\pi_0)$ governs the width of the MONITOR zone, where π_1 and π_0 are the fire and non-fire prevalences.

5. Experiments

5.1. Setup

We evaluate three model families under three decision frameworks: **(i)** standard thresholding at $\hat{p} \ge 0.5$, **(ii)** CRC with $\alpha = 0.05$, and **(iii)** three-way CRC with cost-weighted calibration. All thresholds are computed on the held-out calibration set and applied unchanged to the test set.

Metrics. Our *primary* metrics are: **coverage** ($1-\text{FNR}$), the fraction of true fire pixels correctly flagged (target: $\ge$95%); and **set size**, the fraction of all valid pixels flagged as fire (smaller is better, given safety is met). *Secondary* metrics include AUROC, precision, F1, IoU, AUPRC, and three-way zone statistics.

5.2. Main Results

Table 1 reveals four key findings.

Standard thresholds are unsafe. At $\hat{p} \ge 0.5$, LightGBM captures only 7.2% of fire pixels, U-Net 39.0%, and ResGNN-UNet 51.5%. Even at the ResGNN-UNet's F1-optimal threshold ($\hat{p} \ge 0.6$), 65.8% of fires are missed. The $\sim$95% non-fire class dominance concentrates probability mass near zero, making any conventional threshold catastrophically unsafe (Figure 1). The FNR sweep (Figure 2) confirms this: safe FNR is achievable only at thresholds far below 0.5.

Table 1. Test-set results across three models and three decision frameworks. CRC targets $\mathrm{FNR} \leq 0.05$. Three-way CRC uses c_{fn}=5, c_{fp}=1, $\rho \in [0.9, 1.1]$. ResGNN-UNet rows report recall at fixed thresholds; CRC was not applied because the model was trained externally and its calibration-set probabilities were not available in our reproducible pipeline.

Method	Params	$\hat{\lambda}$	Cov.↑	FNR↓	Set↓	AUC↑
LGBM ($p\geq.5$)	—	.500	.072	.928	.001	.854
LGBM+CRC	—	.003	.941	.059	.626	.854
LGBM+3way	—	[0,.03]	1.00	.000	.040	.854
UNet ($p\geq.5$)	470K	.500	.390	.610	.007	.969
UNet+CRC	**470K**	**.002**	**.947**	**.053**	**.149**	**.969**
UNet+3way	470K	[0,.02]	1.00	.000	.051	.969
ResGNN ($p\geq.5$)	229K	.500	.515†	.485	—	.951
ResGNN ($p\geq.6$)	229K	.600	.342†	.658	—	.951

†Recall at F1-optimal threshold on validation set.

Table 2. Three-way zone fractions on the test set. The SAFE zone collapses because $B_{\mathrm{pw}} = \max(c_{\mathrm{fn}}\pi_1, c_{\mathrm{fp}}\pi_0) = 0.95$ is dominated by $c_{\mathrm{fp}}\pi_0$ under $\sim$5% fire prevalence.

Model	SAFE	MONITOR	EVACUATE	B_{pw}
LightGBM	0.0%	95.96%	4.04%	0.988
U-Net	0.0%	94.90%	5.10%	0.987

Table 3. Model training summary. All models trained from scratch on identical NDWS splits.

Model	Params	Epochs	Optim	Val Loss	AUC
LightGBM	—	100 iter	GBM	—	.854
Tiny U-Net	470K	50	AdamW	0.048	.969
ResGNN-UNet	229K	12†	AdamW	0.170	.951

†Early-stopped from 25 (patience=7, best epoch 5).

CRC delivers safety; model quality delivers efficiency. CRC raises both primary models to $\sim$95% coverage, meeting the $\mathrm{FNR} \leq 0.05$ target. However, the cost differs substantially (Figure 4): LightGBM must flag 62.6% of all pixels, while U-Net flags only 14.9%—a **4.2× reduction** in evacuation zone size. This gap is explained by the AUROC difference (0.854 vs. 0.969): better fire/non-fire separation allows CRC to set a higher $\hat{\lambda}$ while maintaining the same safety guarantee. Figure 8 (Appendix) visualizes this safety–efficiency decomposition across all six configurations.

Architectural complexity $\neq$ better separation. The ResGNN-UNet achieves lower AUROC (0.951) than the Tiny U-Net (0.969) despite its more complex architecture (graph attention bottleneck). On 64×64 patches, the U-Net's receptive field already covers the full spatial extent; the GNN bottleneck adds overhead without improving separation. This reinforces our central claim: *a simple model with CRC outperforms a complex model without it*.

Per-pixel risk gradients. The U-Net produces smooth probability gradients where fire risk decays with distance from fire centers (Figure 6). The CRC threshold ($\hat{\lambda} = 0.002$) captures the entire "tail" of low-probability but real fire pixels that conventional thresholds miss, yielding an interpretable risk map where the CRC boundary defines the guaranteed-safe evacuation zone (Figure 3).

5.3. Before and After CRC

Figure 5 presents a multi-sample comparison of fire detection with and without CRC. Without CRC, the standard threshold produces predominantly red error maps—vast regions of missed fire. With CRC ($\hat{\lambda} = 0.002$), the maps turn green (correctly detected) with some blue (over-alerts), and the three-way CRC provides an interpretable MONITOR/EVACUATE policy.

5.4. Risk Gradient Analysis

Figure 6 visualizes the per-pixel probability landscape. Rather than binary predictions, the U-Net produces a continuous risk gradient where each pixel carries a calibrated fire probability. The CRC threshold contour and the EVACUATE boundary provide principled decision surfaces overlaid on this risk field.

5.5. Three-Way Zone Analysis

Table 2 shows the zone breakdown. Three-way CRC achieves 100% coverage with small EVACUATE zones (4–5%), but the SAFE zone collapses entirely ($\lambda_{\min} = 0$). With $\sim$5% fire prevalence:

$$B_{\mathrm{pw}} = \max(5 \times 0.05,\ 1 \times 0.95) = 0.95,$$

dominated by the majority-class term $c_{\mathrm{fp}}\pi_0$. The resulting shift correction exceeds the base CRC threshold ($\sim$0.003), pushing $\lambda_{\min}$ below zero and producing a maximally conservative policy: all non-evacuated pixels are routed to human review. This reveals a fundamental limitation of the prevalence-weighted bound in rare-event regimes where $\pi_0 \gg \pi_1$.

5.6. Training Details

Table 3 summarizes the training investment. The Tiny U-Net achieves the best AUROC (0.969) with a standard convolutional architecture and no graph-processing dependencies. The ResGNN-UNet early-stopped at epoch 12 (best epoch 5, val loss 0.170 vs. train loss 0.12), exhibiting overfitting that suggests the graph attention bottleneck does not generalize well at this patch scale. Figure 7 shows the U-Net's smooth convergence with minimal overfitting.

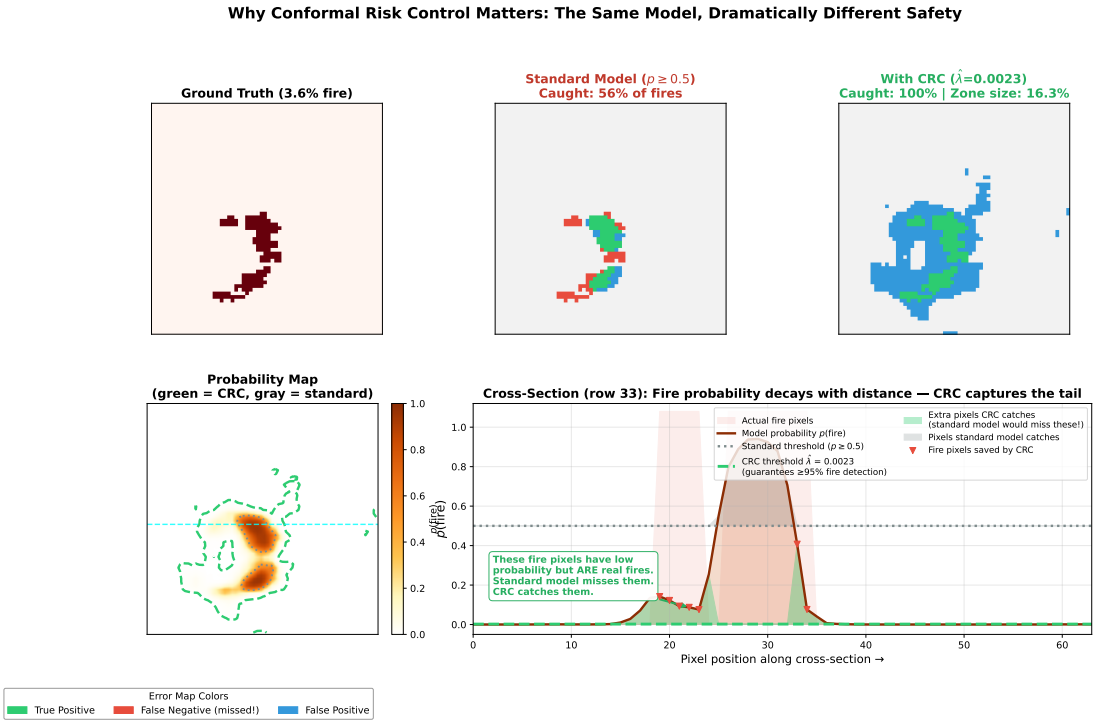

Figure 3. Same U-Net, different thresholds. Standard thresholding ($\hat{p} \geq 0.5$) catches 56% of fires; CRC ($\hat{\lambda} = 0.002$) catches 100%. Bottom: probability cross-section showing how CRC captures low-probability fire pixels. Green shading marks pixels saved by CRC.

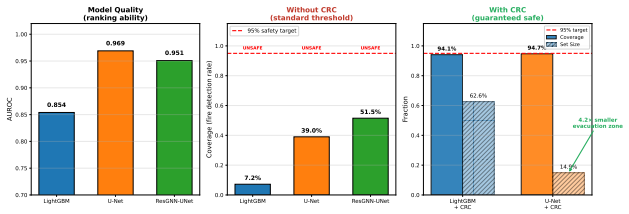

Figure 4. Three-model comparison. **Left:** AUROC. **Center:** without CRC, no model reaches 95% coverage. **Right:** with CRC, both meet the safety target; U-Net achieves 4.2× smaller evacuation zones.

6. Discussion

Separation of safety and efficiency. Our central result is that CRC *cleanly separates* safety from efficiency. All models achieve the same FNR guarantee; what varies is the evacuation zone size. This decomposition has practical significance: teams can invest in better models to reduce evacuation costs without compromising or re-validating safety properties.

Diminishing returns from complexity. The ResGNN-UNet, despite its graph attention bottleneck designed to model fire propagation, achieves lower AUROC than the Tiny U-Net. On 64×64 patches, the U-Net's receptive field spans the full tile; GNN message-passing over an 8-connected grid provides no information that spatial convolutions cannot capture. This echoes broader findings that inductive bias alignment matters more than raw model capacity (Ronneberger et al., 2015).

Limitations of three-way CRC under imbalance. The three-way framework is conceptually appealing—routing uncertain pixels to human review rather than binary decisions. However, under extreme imbalance ($\sim$5% fire), B_{pw} is dominated by the majority-class cost, collapsing the SAFE zone. Future work could explore: **(i)** tighter bounds for rare-event settings, **(ii)** hybrid approaches using separate precision-based thresholds for SAFE, or **(iii)** regional calibration where local prevalence is higher.

Limitations and assumptions. CRC's guarantee requires exchangeability between calibration and test data, which may be violated under distributional shift (e.g., novel fire regimes, climate change). Our analysis uses a single dataset (NDWS) with US wildfire events; generalization to other geographies requires further validation. The three models represent a limited architectural sweep; transformer-based architectures may yield different efficiency–complexity trade-offs.

Future work. Our results open several directions that we leave to future investigation.

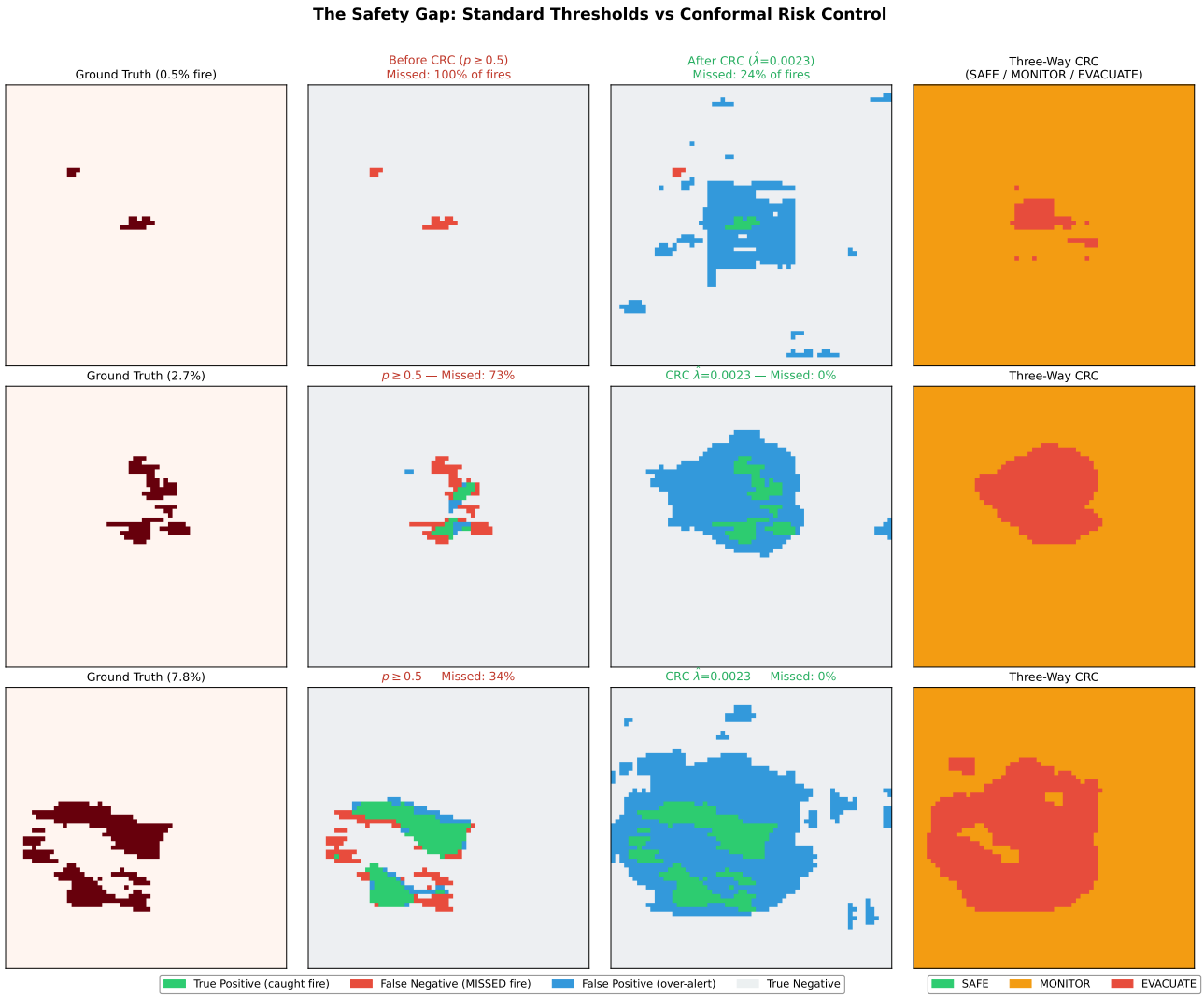

Figure 5. The safety gap across three test samples (0.5%, 2.7%, 7.8% fire). **Col. 1:** ground truth. **Col. 2:** standard threshold—missed fire in red (100%, 73%, 34% missed). **Col. 3:** CRC threshold—improved detection (green). **Col. 4:** three-way zones (SAFE/MONITOR/EVACUATE).

Class-conditional three-way CRC. The three-way zone collapse under extreme imbalance (Section 5.5) stems from coupling both error types through a single prevalence-weighted bound. A natural fix is to calibrate each zone boundary independently: $\lambda_{\max}$ via FNR-CRC on positive pixels and $\lambda_{\min}$ via FPR-CRC on negative pixels. Because the negative class has abundant calibration data ($\sim$95% of pixels), this approach should produce a non-degenerate SAFE zone regardless of class imbalance—resolving the primary limitation of our current framework. We did not pursue this here because the theoretical guarantees require a careful joint analysis of two simultaneous conformal procedures, which is beyond the scope of this comparative study.

Image-level CRC. Our current calibration pools all pixels across calibration images, treating each pixel as an independent sample. Pixels within the same 64×64 patch are spatially correlated, which formally violates the exchangeability assumption underlying CRC. An image-level formulation—defining the loss as per-image FNR and applying CRC to the n=2,781 image-level losses—would restore theoretical validity at the cost of a smaller effective sample size. Comparing pixel-pooled and image-level CRC would quantify whether the practical convenience of pooling degrades the guarantee, and would produce per-image FNR distributions that are more operationally meaningful than aggregate statistics.

Formalizing the AUROC–efficiency relationship. Our central finding—that model discrimination determines evacuation zone size under a fixed safety guarantee—is currently empirical. Under a bi-normal score model, the CRC set size

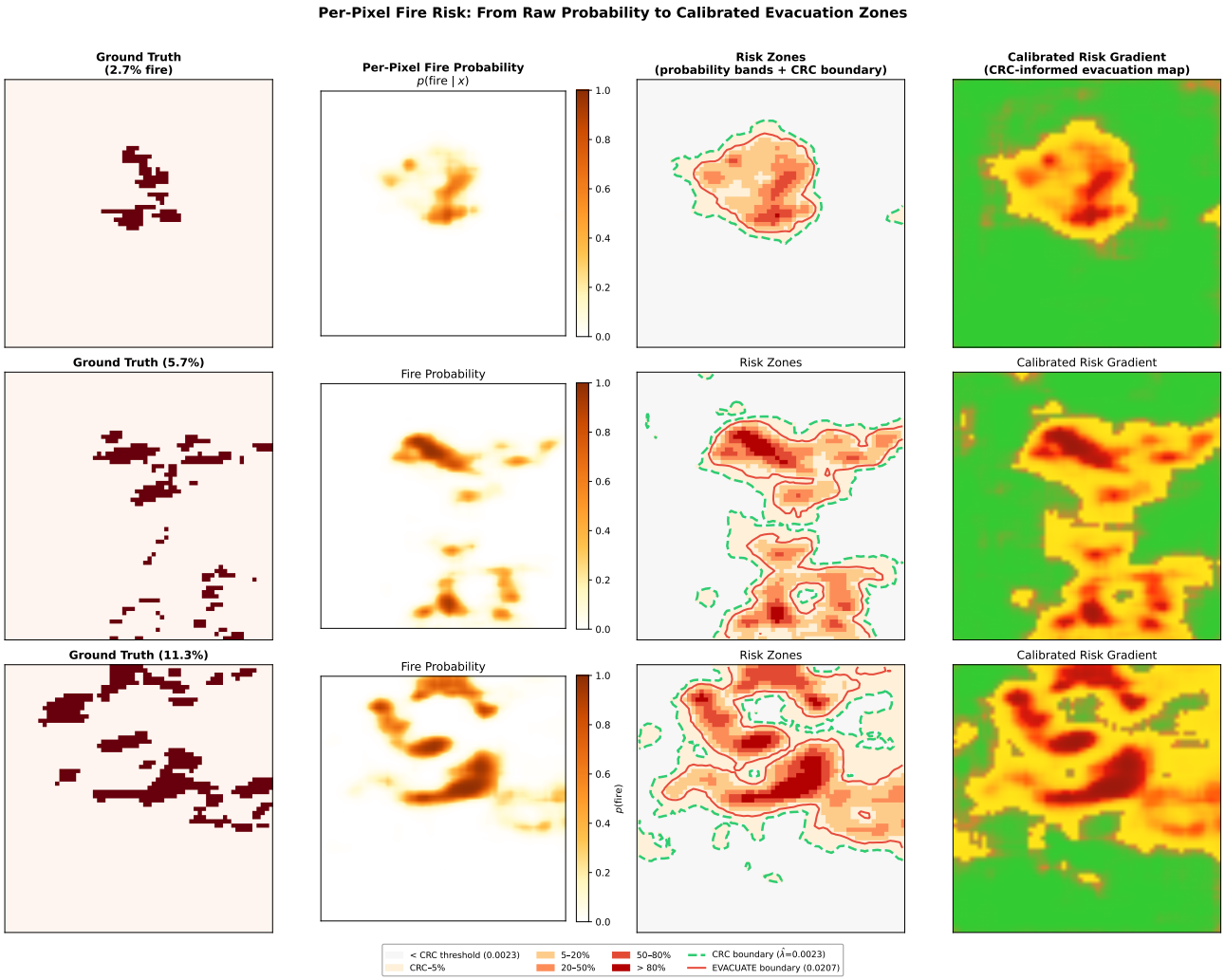

Figure 6. Per-pixel fire risk gradient. **Col. 1:** ground truth. **Col. 2:** continuous probability heatmap. **Col. 3:** probability bands with CRC contour (green dashed) and EVACUATE boundary (red). **Col. 4:** calibrated risk gradient for operational mapping.

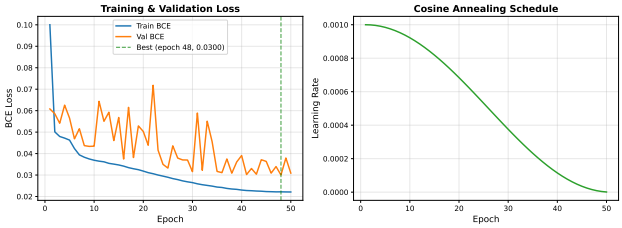

Figure 7. Tiny U-Net training curves: BCE loss and cosine annealing LR over 50 epochs. Smooth convergence with minimal train–val gap.

decomposes as $S(\alpha) = \pi_1(1-\alpha) + \pi_0(1 - F_0(F_1^{-1}(\alpha)))$, where F_0, F_1 are the class-conditional score CDFs. This would yield a closed-form prediction of evacuation cost from AUROC alone, enabling practitioners to answer: *what discrimination level is needed for a target zone size?* We leave the formal statement and empirical validation of this relationship to future work.

Temporal and geographic shift. CRC assumes exchangeability between calibration and test data. Wildfire behavior varies across seasons, years, and geographies due to climate change, drought cycles, and regional fuel loads—creating natural label shift that may violate this assumption. The shift-aware recalibration framework of Angelopoulos et al. (2022) could address this, but validating it requires temporally stratified splits and multi-region datasets that the current NDWS benchmark does not cleanly support. Evaluating CRC robustness under realistic temporal shift is critical for any operational deployment.

7. Conclusion

We have demonstrated that conformal risk control transforms wildfire spread prediction from an accuracy-optimization problem into a safety-guaranteed decision system. Across three model families, our experiments establish that:

1. **No model is safe without CRC.** Standard thresholding captures only 7–52% of fires.
2. **CRC guarantees safety regardless of model.** Both calibrated models achieve $\geq 94\%$ coverage on held-out data.
3. **Model quality determines efficiency, not safety.** The Tiny U-Net achieves $4.2\times$ tighter evacuation zones than LightGBM under the same CRC guarantee.
4. **Complexity without separation is waste.** A graph-augmented model underperforms a standard U-Net in AUROC despite its more complex architecture.

For practitioners: a well-trained simple model wrapped in CRC provides both stronger safety guarantees and tighter evacuation zones than a complex model evaluated on F1 alone. The few lines of conformal calibration code deliver what no amount of architectural sophistication can—a mathematical guarantee that $\geq 95\%$ of fires will be detected.

Impact Statement

This work targets safer wildfire evacuation decision support by explicitly controlling false negatives. Potential positive impact includes better risk-aware public safety planning. Potential negative impact includes over-reliance on automated recommendations; we mitigate this through the three-way framework that routes uncertain pixels to human review.

Software and Data

All code, trained models, and evaluation pipelines are maintained in a reproducible repository with deterministic data splits and CLI entry points for each phase. The dataset is NDWS (Haddad, 2024), subject to its original terms of use.

References

Amodei, D., Olah, C., Steinhardt, J., Christiano, P., Schulman, J., and Mané, D. Concrete problems in AI safety. *arXiv preprint arXiv:1606.06565*, 2016.

Angelopoulos, A. N. and Bates, S. A gentle introduction to conformal prediction and distribution-free uncertainty quantification. *arXiv preprint arXiv:2107.07511*, 2021.

Angelopoulos, A. N., Bates, S., Candès, E. J., and Jordan, M. I. Conformal risk control. *arXiv preprint arXiv:2208.02814*, 2022.

Barber, R. F., Candès, E. J., Ramdas, A., and Tibshirani, R. J. Predictive inference with the jackknife+. *The Annals of Statistics*, 49(1):486–507, 2021.

Guo, C., Pleiss, G., Sun, Y., and Weinberger, K. Q. On calibration of modern neural networks. In *International Conference on Machine Learning*, 2017.

Haddad, F. Next day wildfire spread dataset. Kaggle, 2024. Accessed 2026-03-17; dataset identifier: fantineh/next-day-wildfire-spread.

Huot, F., Hu, R. L., Goyal, N., Sankaranarayanan, T., Angelopoulos, A. N., et al. Next day wildfire spread: A machine learning dataset for predicting wildfire spreading from remote-sensing data. *IEEE Transactions on Geoscience and Remote Sensing*, 60:1–13, 2022.

Jain, P., Coogan, S. C., Subramanian, S. G., Crowley, M., Taylor, S., and Flannigan, M. D. A review of machine learning applications in wildfire science and management. *Environmental Reviews*, 28:478–505, 2020.

Ke, G., Meng, Q., Finley, T., Wang, T., Chen, W., Ma, W., Ye, Q., and Liu, T.-Y. Lightgbm: A highly efficient gradient boosting decision tree. In *Advances in Neural Information Processing Systems*, 2017.

Kipf, T. N. and Welling, M. Semi-supervised classification with graph convolutional networks. In *International Conference on Learning Representations*, 2017.

Platt, J. C. Probabilistic outputs for support vector machines and comparisons to regularized likelihood methods. In *Advances in Large Margin Classifiers*. MIT Press, 1999.

Radke, D., Hessler, A., and Ellsworth, D. Firecast: Leveraging deep learning to predict wildfire spread. *Proceedings of the International Joint Conference on Artificial Intelligence*, 2019.

Romano, Y., Sesia, M., and Candès, E. Classification with valid and adaptive coverage. *Advances in Neural Information Processing Systems*, 2020.

Ronneberger, O., Fischer, P., and Brox, T. U-net: Convolutional networks for biomedical image segmentation. In *Medical Image Computing and Computer-Assisted Intervention (MICCAI)*, 2015.

Shafer, G. and Vovk, V. A tutorial on conformal prediction. *Journal of Machine Learning Research*, 9:371–421, 2008.

Veličković, P., Cucurull, G., Casanova, A., Romero, A., Liò, P., and Bengio, Y. Graph attention networks. In *International Conference on Learning Representations*, 2018.

Vovk, V., Gammerman, A., and Shafer, G. *Algorithmic Learning in a Random World*. Springer, 2005.

Zhao, X., Xu, Y., Lovreglio, R., Kuligowski, E., Nilsson, D., Cova, T. J., Wu, A., and Yan, X. Estimating wildfire evacuation decision and departure timing using large-scale GPS data. *Transportation Research Part D: Transport and Environment*, 107:103277, 2022.

A. Additional Figures

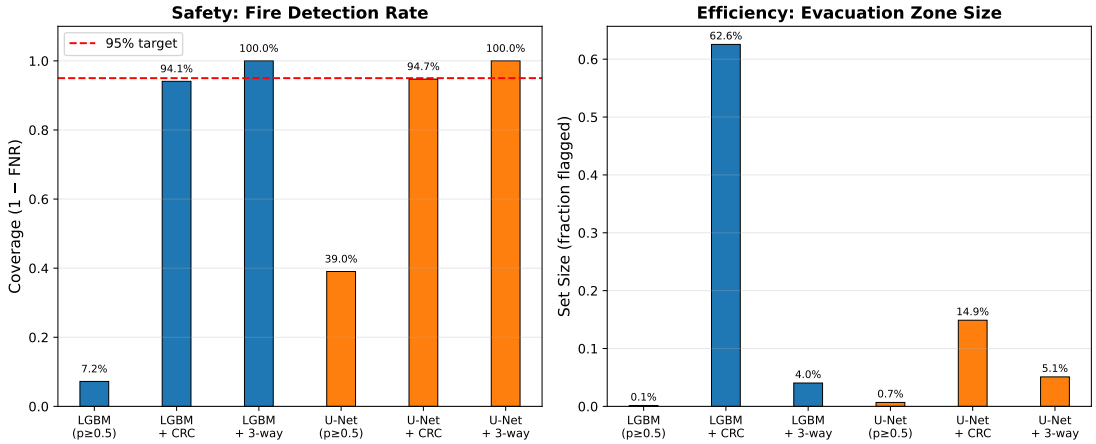

Figure 8. Safety vs. efficiency for all six configurations. **Top:** coverage (safety)—only CRC and three-way methods achieve $\geq 95\%$. **Bottom:** set size (efficiency)—U-Net+CRC is the optimal operating point.

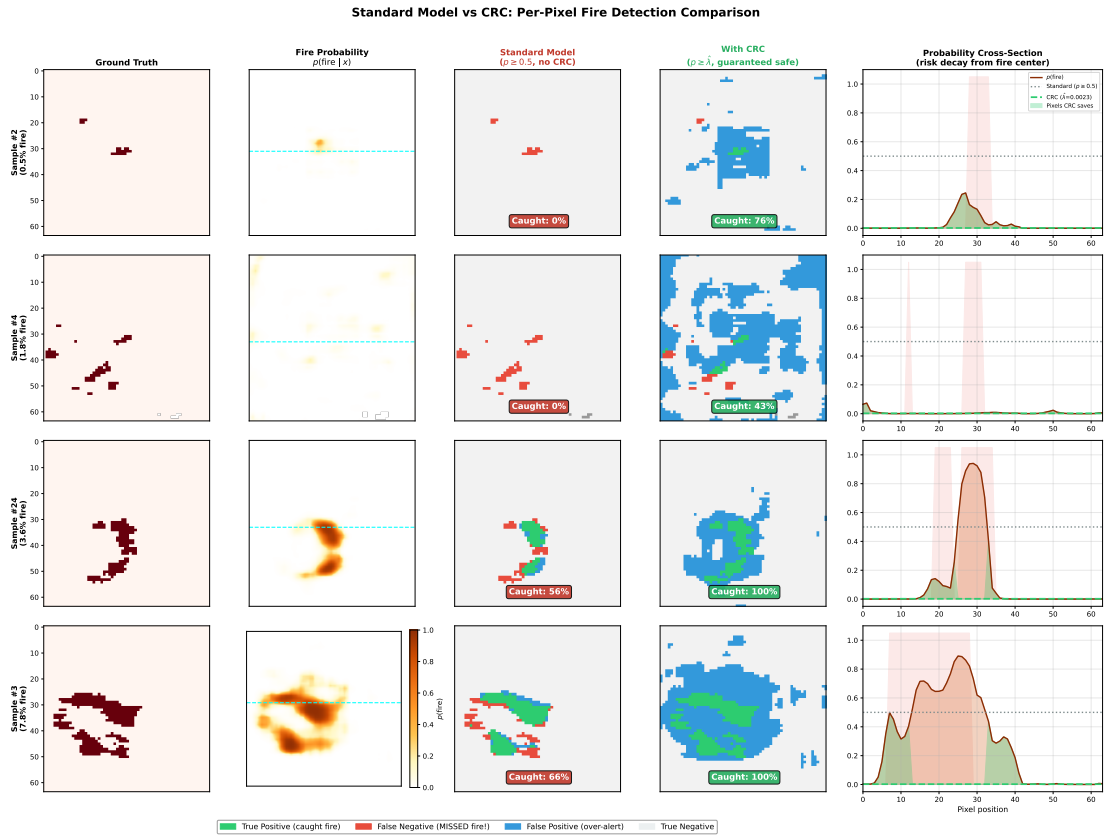

Figure 9. Bare model vs. CRC across four test samples. Fire catch rate improves from 8%→98%, 0%→83%, 56%→100%, 48%→100%.

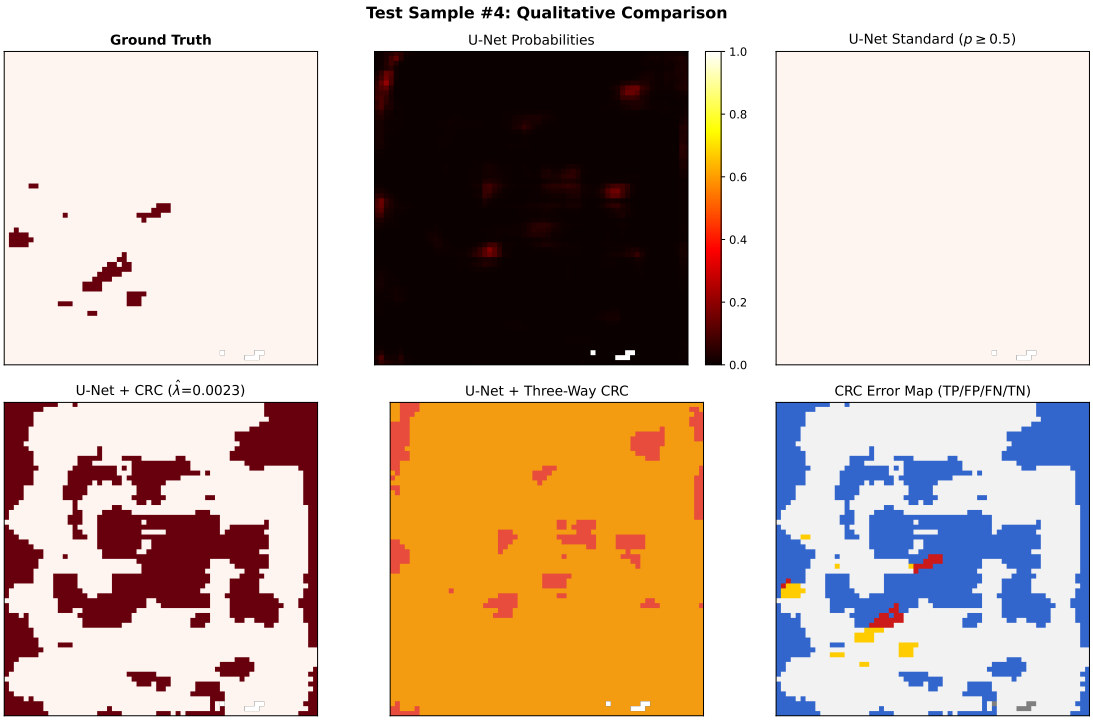

Figure 10. Qualitative prediction maps: ground truth, probabilities, standard prediction, CRC prediction, three-way zones, and error analysis.